

Which of the following carbon-carbon bonds in the molecules below can freely rotate?

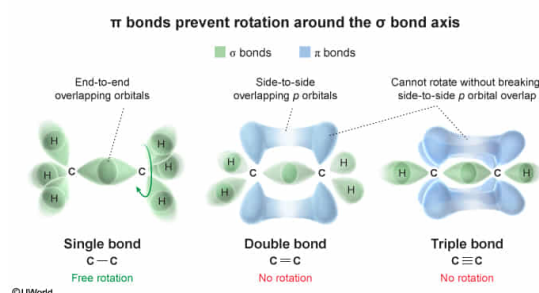
- I. $\text{H}_3\text{C}-\text{CH}_3$
- II. $\text{H}_2\text{C}=\text{CH}_2$
- III. $\text{HC}\equiv\text{CH}$

- ✓ ☒ A. I only [96%]
☐ B. II only
☐ C. I and III only
☐ D. II and III only [1%]

Correct

96% answered correctly

Explanation:



Covalent bonds between atoms are formed by **sharing electrons**, which is made possible through the overlap of the **atomic orbitals** from each atom. Covalent bonds made by the **end-to-end overlap** of atomic orbitals are called **sigma (σ) bonds** whereas covalent bonds made by the **side-to-side overlap** of p orbitals are called **pi (π) bonds**.

Although some atoms only participate in single covalent bonding (by a σ bond), other atoms are capable of participating in **multiple bonds** (double or triple bonds) by the addition of one or more π bonds. In this way, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds.

Because σ bonds are made by an end-to-end orbital overlap, **free rotation** around the bond between the two σ -bonded atoms (such as a $\text{C}-\text{C}$ single bond) can be achieved while maintaining the orbital overlap. Accordingly, substituents such as methyl groups ($-\text{CH}_3$) can fully rotate 360° around the carbon σ -bond axis (**Number I**).

Conversely, free rotation around the bond between two π -bonded atoms (such as in a $\text{C}=\text{C}$ double bond or a $\text{C}\equiv\text{C}$ triple bond) is not possible due to side-to-side orbital overlap. Rotation would disrupt the p orbital overlap between the atoms and break the π bond; as such, π bonds lock bond rotation. Therefore, the hydrogen atoms bonded in a $=\text{CH}_2$ group cannot freely rotate around the double bond axis (**Number II**). Likewise, a $\text{C}\equiv\text{C}$ bond cannot have rotation between the carbon atoms (**Number III**).

Educational objective:

Covalent bonds are formed by sharing electrons between atoms through the overlap of

atomic orbitals in an end-to-end or side-to-side configuration. Multiple bonds formed by s and p orbitals consist of one σ bond and one or more π bonds. Free rotation around a bond is possible with σ bonds (single bonds) but is prevented by the structure of π bonds (double and triple bonds).

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